# ISMAEL OMAR LEONIDAS HURTADO VARGAS

*Senior Full Stack Engineer · Artificial Intelligence / LLM*

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## PROFILE

Senior Full Stack Engineer with 10+ years building end-to-end solutions with React, Node.js, TypeScript, Python and microservices architectures (NestJS). Specialized in AI-powered applications (self-hosted LLMs, chatbots, RAG, OCR, computer vision), payment (Niubiz, Izipay, Culqi), SUNAT e-invoicing and SAP integrations, and data analysis. I apply SOLID and clean architecture (Hexagonal, DDD) and deploy with Docker, Kubernetes (k3s), CI/CD and AWS. Bachelor's degree in Statistical Engineering, completed M.Sc. coursework in Systems and Computer Engineering with a major in Software Engineering (UNMSM), and current M.Sc. student in AI (UCV).

## PROFESSIONAL EXPERIENCE

### Sicred Informática Ltda

*Mortgage lending · Mexico*

*Senior Full Stack Developer · Remote* — January 2026 to September 2026

- Developed the credit-agent platform: microservices, a mobile app and a web frontend for credit-agent management and operations (NestJS, Fastify, Prisma, React Native, Expo, React, RSuite, Tailwind CSS, hexagonal architecture, DDD).
- Defined the platform architecture: layers, technical standards, CI/CD and security (hexagonal architecture, DDD, Clean Architecture).
- Developed DAEFlow, the FOVISSSTE mortgage case origination platform: stage-based workflow, notaries, market dashboards, maps and comparison of financial institutions (NestJS, Fastify, Prisma, React, RSuite, Leaflet, ApexCharts).
- Built the on-site Construction Verification system: microservices for RUV orders, progress, logbook and INFONAVIT report formats, plus a mobile app with geolocation, maps and real-time alerts (NestJS, Prisma, React Native, Expo, SQLite, SocketCluster).
- Built the legal-document comparison platform: in-house OCR and a self-hosted AI engine that contrasts documents and summarizes case files in a chat (Python, FastAPI, PyMuPDF, Tesseract, Ollama).
- Deployed the Construction Verification system on Kubernetes: a test cluster with its 5 microservices and frontend, and automated deployment from the repository (k3s, Kustomize, Traefik, GitOps).
- Automated QA and production CI/CD: semantic versioning, secret scanning and security with WAF, rate limiting and captcha (GitHub Actions, Docker Buildx, gitleaks, Cloudflare, Helmet).

### Municipalidad Metropolitana de Lima

*Public sector · Lima, Peru*

*Senior Programmer Analyst · Hybrid* — July 2023 to December 2025

- Was part of the team that developed SIAM: a microservices ecosystem with per-module CI/CD (Node.js, TypeScript, Docker, GitHub Actions, hexagonal architecture, DDD).
- Developed modules of the Electronic Enforcement System: REST APIs, a geospatial viewer and video streaming for photo and video ticket evidence (Node.js, Express, JWT, deck.gl, Leaflet).
- Developed the Business Operating Licenses module: zoning-based business activities, risk and ITSE, advertising permits and license issuance with digital signature (NestJS, hexagonal architecture, DDD, Firma Perú).
- Built LimaBot, a citizen-support chatbot: questions about procedures, services and TUPA (OpenAI, Node.js, TypeScript, React).
- Integrated the government PIDE interoperability services: secure data exchange with RENIEC, SAT and Immigration (Node.js, TypeScript, REST).

### Urban Transport Authority (ATU)

*Public sector · Lima, Peru*

*Programmer Analyst · Hybrid* — January 2021 to January 2023

- Implemented the tickets module of the Activity Registry System (SRA): registration and management of infractions through REST APIs (Node.js, Express, SOLID).
- Developed the TUPA module of the Payment Gateway (pasarela.atu.gob.pe): procedure catalog, debt lookup, online payment, QR-coded receipt, user registration and reCAPTCHA (Node.js, TypeScript, JWT, SocketCluster, Niubiz).
- Created the Symptom Record and SISATU modules: access for authorized personnel and visitors, and payments and debt inquiry (Node.js, Express, SQL Server).
- Designed the warehouse and work-order module of the Operational Management System and maintained SISCON: warehouse management, work orders and recruitment calls (Node.js, Express, SQL Server).

### Cobra Perú

*Telecommunications · Lima, Peru*

*Systems Analyst · On-site* — December 2014 to December 2020

- Led the selection and training of the web team and implemented the GIA system for the Andean Zone: administration management integrated with the Navision ERP for purchasing, pre-orders, invoices and assets (Node.js, Express, SocketCluster, SQL Server, MongoDB).
- Built the GIA cross-cutting modules: document management (DocuCobra), users, profiles and per-option permissions, and PDF and Excel reports (Node.js, Express, pdfmake, excel4node, SQL Server, MongoDB).

- Developed production and quality KPI modules and reports: stored procedures, automated email delivery and database backups (JavaScript, Node.js, C#, SQL Server).
- Built dashboards and KPIs for senior management: production and quality indicators for client Telefónica del Perú (Power BI, Qlik Sense).

## PROJECTS

- **Exandal Holding Group** (agro-industry): extended its quality system integrated with SAP HANA and industrial scales, and built the Tickets module, ExcoBot (natural-language queries over SAP) and a RAG chatbot for quality and food safety (Node.js, SAP HANA, OpenAI, RAG).
- **BizPeru** (ISP and WiFi): built its management system with SUNAT electronic invoicing, card and Yape payments, an AI WhatsApp assistant and network automation (Node.js, MySQL, React, Nubefact, Izipay, Culqi, OpenAI, Meta Cloud API, MikroTik, SmartOLT).
- **Tawa Telcom** (ISP · tawatelcom.com.pe): built the internet provider's website with an AI assistant, an integrated complaints book and SEO with structured data (Next.js 16, Tailwind CSS 4, TypeScript).
- **Chatt World** (personal project · chatt-world.com): built a group chat with automatic translation into each reader's language, a PWA with notifications and a multilingual interface (Fastify, WebSockets, PostgreSQL, React, Zustand, Web Push).
- **Notes** (personal project · ismaelhv.com/notes): built a notes and tasks app with nested notebooks, a "Today" view, a journal, relevance search, an AI writer and link sharing (NestJS, Fastify, Prisma, PostgreSQL, React, TipTap, OpenAI).
- **waHub** (personal project · ismaelhv.com/wahub): built a platform that connects WhatsApp numbers to AI agents through a visual flow editor (NestJS, Fastify, Baileys, Prisma, PostgreSQL, Next.js, React Flow, OpenAI).
- **Atalaya** (M.Sc. in AI · computer vision): built personal protective equipment detection on video with RTSP cameras, alerts and a web dashboard (YOLO11n, Ultralytics, PyTorch, OpenCV, FastAPI).
- **FastLink and DevTools** (personal projects · ismaelhv.com): built a URL shortener with anti-bot protection and in-browser developer utilities that never upload data (Astro, React, Tailwind CSS, Cloudflare Turnstile).

## EDUCATION

**M.Sc. in Artificial Intelligence** *In progress · since 2026*
*Universidad César Vallejo*

**M.Sc. in Systems and Computer Engineering** *Coursework completed · 2020 to 2023*
*Universidad Nacional Mayor de San Marcos · Software Engineering major*

**Statistical Engineering** *Bachelor's degree · 2009 to 2013*
*Universidad Nacional de Trujillo · Top decile*

**Computer Science and IT** *Completed · 2020 to 2025*
*Cibertec Institute*

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Node.js, Python, C#, SQL.

**Frontend and mobile:** React, Next.js, RSuite, Tailwind CSS, React Native, Expo, React Three Fiber (3D).

**Backend:** NestJS, Fastify, Express.js, FastAPI, microservices, REST API, WebSockets, JWT.

**Artificial intelligence:** OpenAI/GPT, Ollama (self-hosted), Langchain, agents, chatbots, RAG, OCR, computer vision (YOLO).

**Data and BI:** Power BI, Qlik Sense, dashboards, KPIs, statistical analysis.

**Databases:** PostgreSQL, SQL Server, MySQL, MariaDB, MongoDB, Prisma.

**DevOps and cloud:** Docker, Docker Compose, GitHub Actions (CI/CD), Kubernetes (k3s, Kustomize), Traefik, Nginx, Cloudflare (WAF, Turnstile), AWS (EC2, S3, IAM), Vercel, Linux, k6 (load testing).

**Architecture and integrations:** SOLID, Clean Code, Hexagonal, DDD, Scrum; SAP, Niubiz, Izipay, Culqi, Nubefact (SUNAT), Firma Perú, PIDE.

## CERTIFICATIONS AND LANGUAGES

**Certifications:** Kubernetes (Udemy, 2026) · SOLID Principles and Clean Code (Udemy, 2024) · Chatbots with OpenAI (Platzi, 2024) · Docker (Udemy, 2024) · React (Udemy, 2023) · TypeScript (Dev-Talles, 2023) · BI Analytics and Big Data Specialization (UNALM, 2020).

**Languages:** Spanish (native) · English (intermediate).